FIZ1951-MUHENDİSLER İÇİN YARIİLETKEN FİZİĞİ ARA-2 FORMÜL KAĞIDI

$$\varepsilon_o = 8.85x10^{-12}\frac{F}{m}$$

$$\mu_o = 4\pi x10^{-7}\,T.A/_m = 1.257x10^{-6}\,H/_m$$

$$q = e = 1.6x10^{-19}C$$

$$\mathrm{h} = 6.63x10^{-34}J.s \quad m_o = 9.11x10^{-31}\,kg$$

$$k_B = 1.38x10^{-23}J/K \quad 1\mathrm{eV} = 1.6x10^{-19}J$$

$$c = 3x10^8\frac{m}{s}$$

$$E = \frac{hc}{\lambda} \qquad E = \frac{1.24\,eV.\mu m}{\lambda(\mu m)} \qquad E = \frac{1240\,eV.nm}{\lambda(nm)}$$

$$\sigma_p = pq\mu_p \qquad \sigma_n = nq\mu_n$$

$$\boxed{\sigma = \frac{ne^2\tau}{m_e^*}} \qquad \vartheta_t = \sqrt{\frac{3k_BT}{m^*}}$$

$$\frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^I}$$

$$H = \frac{NI}{l}$$

$$M = \chi_M H$$

$$\mathrm{B} = \mu_o H + \mu_o M$$

$$H = \frac{B}{\mu}$$

$$\mu = \mu_o\mu_r = \mu_o(1+\chi_M)$$

$$R = \frac{(n_r-1)^2+k^2}{(n_r+1)^2+k^2}$$

$$\vartheta = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \qquad n_r = \frac{c}{\vartheta}$$

$$\alpha(h\nu) = A(h\nu - E_g)^{\gamma}$$

$$I(\mathrm{d}) = I_o\exp(-\alpha d)$$

$$T = (1-R^2)\exp(-\alpha d)$$

$$N_D = n$$

$$W_o = \sqrt{\frac{2\varepsilon_o\varepsilon(V_{bi})}{en}} \qquad (V_{dış} = 0)$$

$$W = \sqrt{\frac{2\varepsilon_o\varepsilon(V_{bi}-V_{dış})}{e\,n}} \qquad (V_{dış} > 0)$$

$$W = \sqrt{\frac{2\varepsilon_o\varepsilon(V_{bi}-(-V_{dış}))}{e\,n}} \qquad (V_{dış} < 0)$$

$$eV_{bi} = E_g + kTLn\frac{N_A\,N_D}{N_C\,N_V}$$

$$\boxed{\mathrm{W}_p{}^2 = \frac{2\varepsilon}{e}V_{bi}\left[\frac{\mathrm{N_D}}{\mathrm{N_A}(\mathrm{N_D}+\mathrm{N_A})}\right]}$$

$$\boxed{\mathrm{W}_n{}^2 = \frac{2\varepsilon}{e}V_{bi}\left[\frac{\mathrm{N}_A}{\mathrm{N_D}(\mathrm{N_D}+\mathrm{N_A})}\right]}$$

$$\boxed{W = \left(\frac{2\varepsilon}{e}V_{bi}\left[\frac{\mathrm{N}_A+\mathrm{N_D}}{(\mathrm{N_D}\mathrm{N_A})}\right]\right)^{1/2}}$$

$$\boxed{E_{max}(x=0) = -\frac{\mathrm{e}}{\varepsilon}\mathrm{N_D}\mathrm{W_n} = -\frac{\mathrm{e}}{\varepsilon}\mathrm{N}_A\mathrm{W}_p}$$

$$I = I_o\left(exp\frac{(qV_{Dış})}{nk_BT} - 1\right) \cong I_o\left(exp\frac{(qV_{Dış})}{nk_BT}\right)$$

$$np = n_i{}^2 \qquad \boxed{V_{bi} = \frac{1}{2}E_{max}\mathrm{W}}$$

$$V_{bi} = \frac{kT}{e}ln\frac{p_pn_n}{n_i{}^2}$$

$$I_\mathrm{D} = I_\mathrm{DSS}\left(1-\frac{V_\mathrm{GS}}{V_\mathrm{GS(off)}}\right)^2$$

$$V_\mathrm{CB} = V_\mathrm{CE} - V_\mathrm{BE}$$

$$V_\mathrm{CE} = V_\mathrm{CC} - V_{R_\mathrm{C}}, \qquad V_{R_\mathrm{C}} = I_\mathrm{C}R_\mathrm{C}$$

$$V_{R_\mathrm{B}} = V_\mathrm{BB} - V_\mathrm{BE}, \qquad V_{R_\mathrm{B}} = I_\mathrm{B}R_\mathrm{B}$$

$$\boldsymbol{I_\mathrm{E} = I_\mathrm{C}+I_\mathrm{B}}$$

$$\boldsymbol{I_\mathrm{E} = \frac{\beta+1}{\beta}I_\mathrm{C}}$$

$$\boldsymbol{I_\mathrm{C} = \alpha I_\mathrm{E}}$$

$$\boldsymbol{\alpha = \frac{\beta}{\beta+1}}$$

$\boldsymbol{I_\mathrm{C} = I_S(e^{V_\mathrm{BB}/V_\mathrm{T}}-1)}$, $\boldsymbol{I_\mathrm{C} \gg I_S}$ şartı ile

$\boldsymbol{I_\mathrm{C} = I_S e^{V_\mathrm{BB}/V_\mathrm{T}}}$ dir.

$$\boldsymbol{I_S = \frac{A_EqD_nn_{p0}}{W}}, \quad \boldsymbol{n_{p0} = n_i^2/N_A}$$

$$\boldsymbol{I_\mathrm{E} = (I_S/\alpha)e^{V_\mathrm{BB}/V_\mathrm{T}}}$$

$$V_\mathrm{T} = \frac{kT}{q}$$

$$V_\mathrm{T} \cong 26\text{ mV (@300K)}$$

$$\beta = \frac{I_\mathrm{C}}{I_\mathrm{B}}, \qquad \boldsymbol{I_\mathrm{B} = \frac{I_\mathrm{C}}{\beta}} \qquad \boldsymbol{\beta = \frac{\alpha}{1-\alpha}}$$

$$\boldsymbol{I_\mathrm{B} = \left(\frac{I_S}{\beta}\right)e^{V_\mathrm{BB}/V_\mathrm{T}}} \qquad \boldsymbol{I_\mathrm{E} = \frac{\beta+1}{\beta}I_S e^{V_\mathrm{BB}/V_\mathrm{T}}}$$

$$\vec{x} = \vec{v}_s\Delta t$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqA\vartheta_s$$

$$J = \frac{I}{A}$$

$$\vec{J} = nq\vec{\vartheta_s}$$

$$V = IR$$

$$R = \rho\frac{l}{A}$$

$$\vec{v}_s = \mu\vec{E} = \frac{q\tau}{m}\vec{E}$$

$$\vec{J} = nq\mu\vec{E}$$

$$\sigma = nq\mu = ne\mu$$

$$q = e$$

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

$$\vec{J} = \sigma\vec{E} \qquad \vec{E} = \frac{V}{l}$$

$$\rho = \frac{1}{ne\mu}$$

$$l = \vartheta_{ort}\tau$$

$$\vartheta_\mathrm{th} = l/\tau$$

$$\sigma = 1/\rho \rightarrow S = 1/R$$

$$\vec{F}_e = q\vec{E} = m_o\vec{a}$$

$$\vec{F}_e = q\vec{E} + q\vec{E}_{Kr} = m^*\vec{a}$$